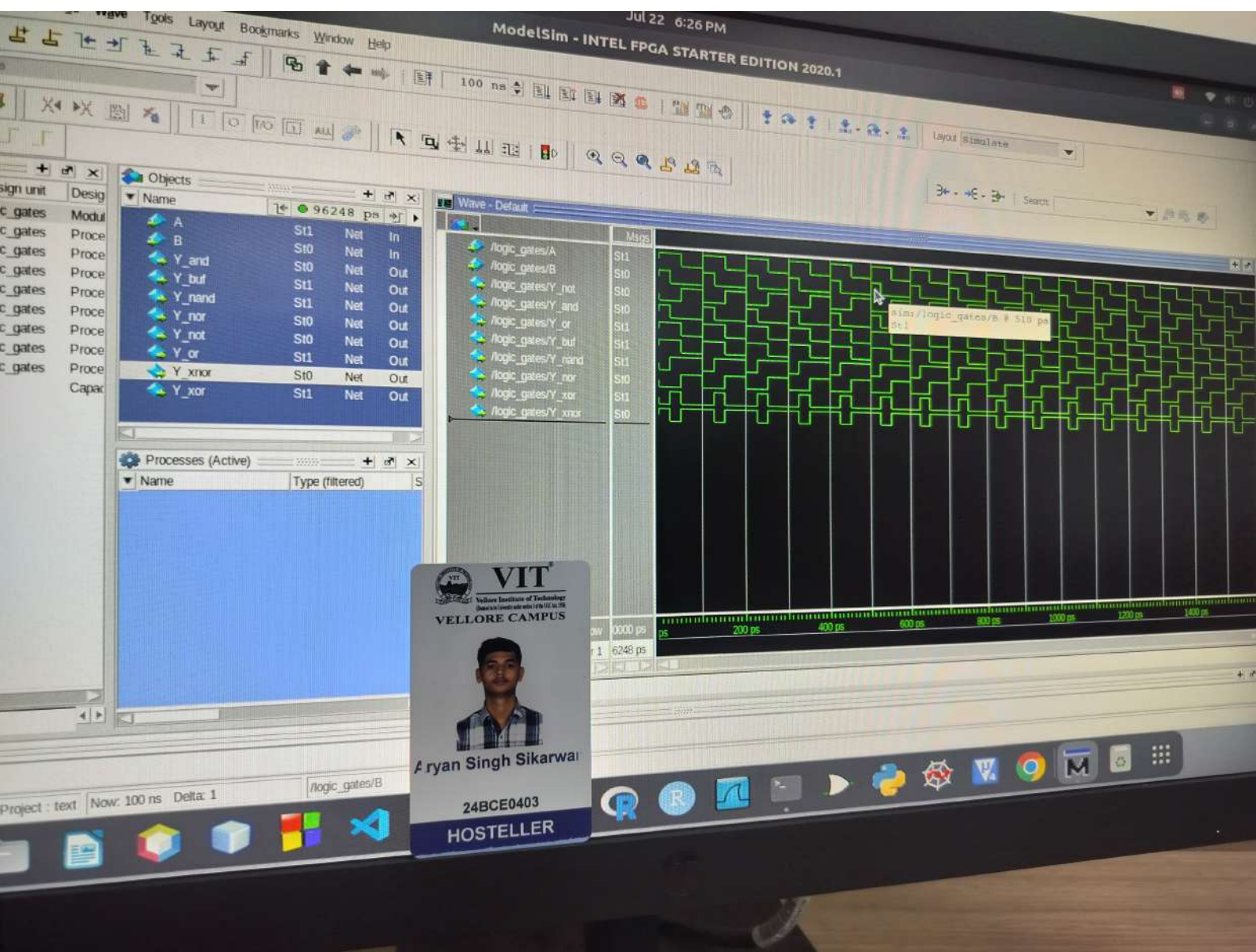




(Que-1) (All logic gates)

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(*) Problem Statement :-

⇒ write verilog code in one file for all gates using gate level modeling and then verify its operation.

Ans

Code :-

```
Module All_gate (Y_and, Y_or, Y_not,  
Y_and, Y_nand, Y_nor, A, B);  
input A, B;
```

```
Output Y_and, Y_or, Y_not, Y_nand  
Y_nor;
```

```
and (Y_and, A, B);
```

```
or (Y_or, A, B);
```

```
not (Y_not, A);
```

```
nand (Y_nand, A, B);
```

```
nor (Y_nor, A, B);
```

```
end module
```

(*) Data flow :-

```
module all_gate (output Y_and, Y_or, Y_n  
Y_nand, Y_nor, input A, B);
```

```
assign Y_and = A & B;
```

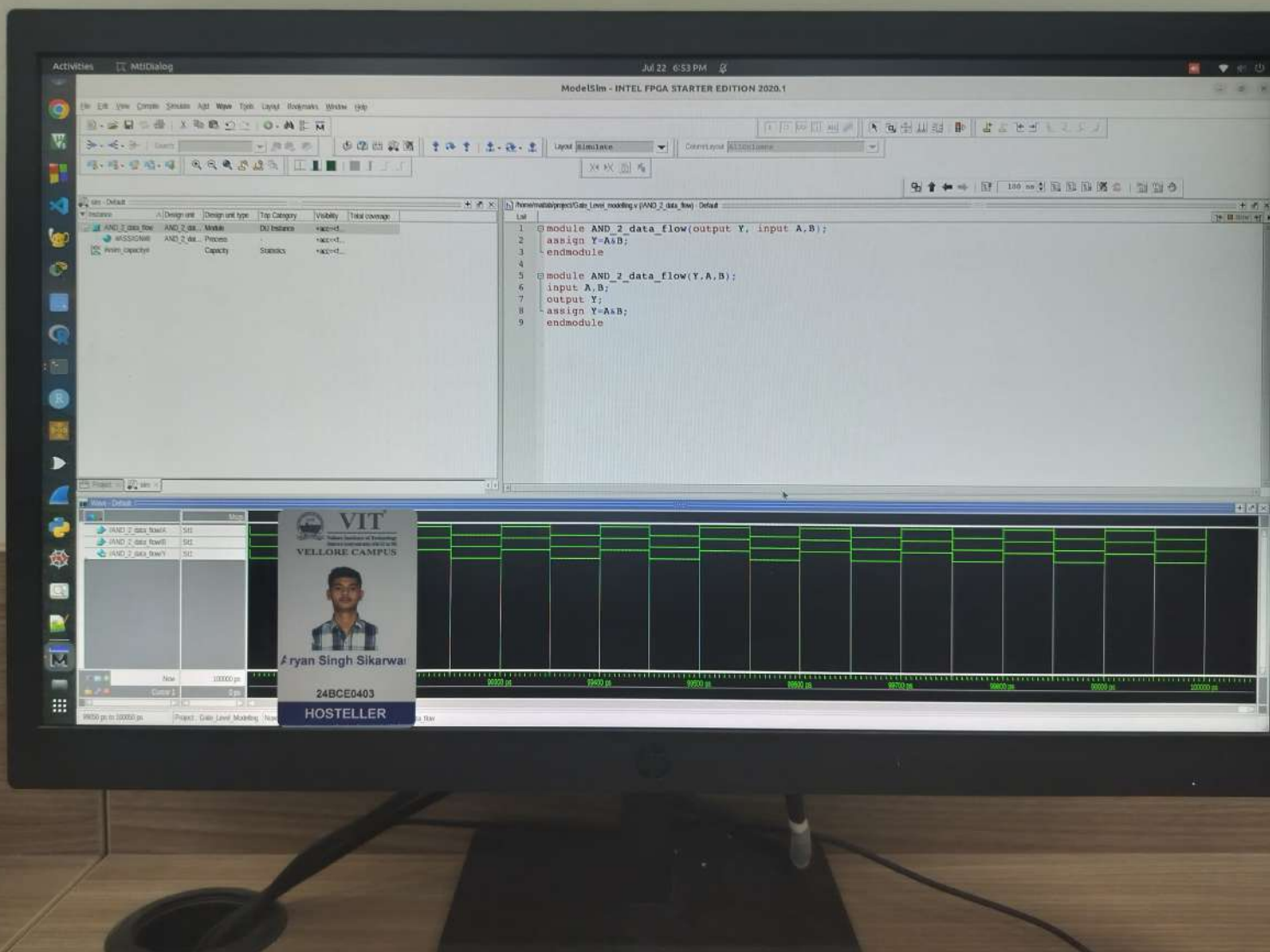
```
assign Y_or = A | B;
```

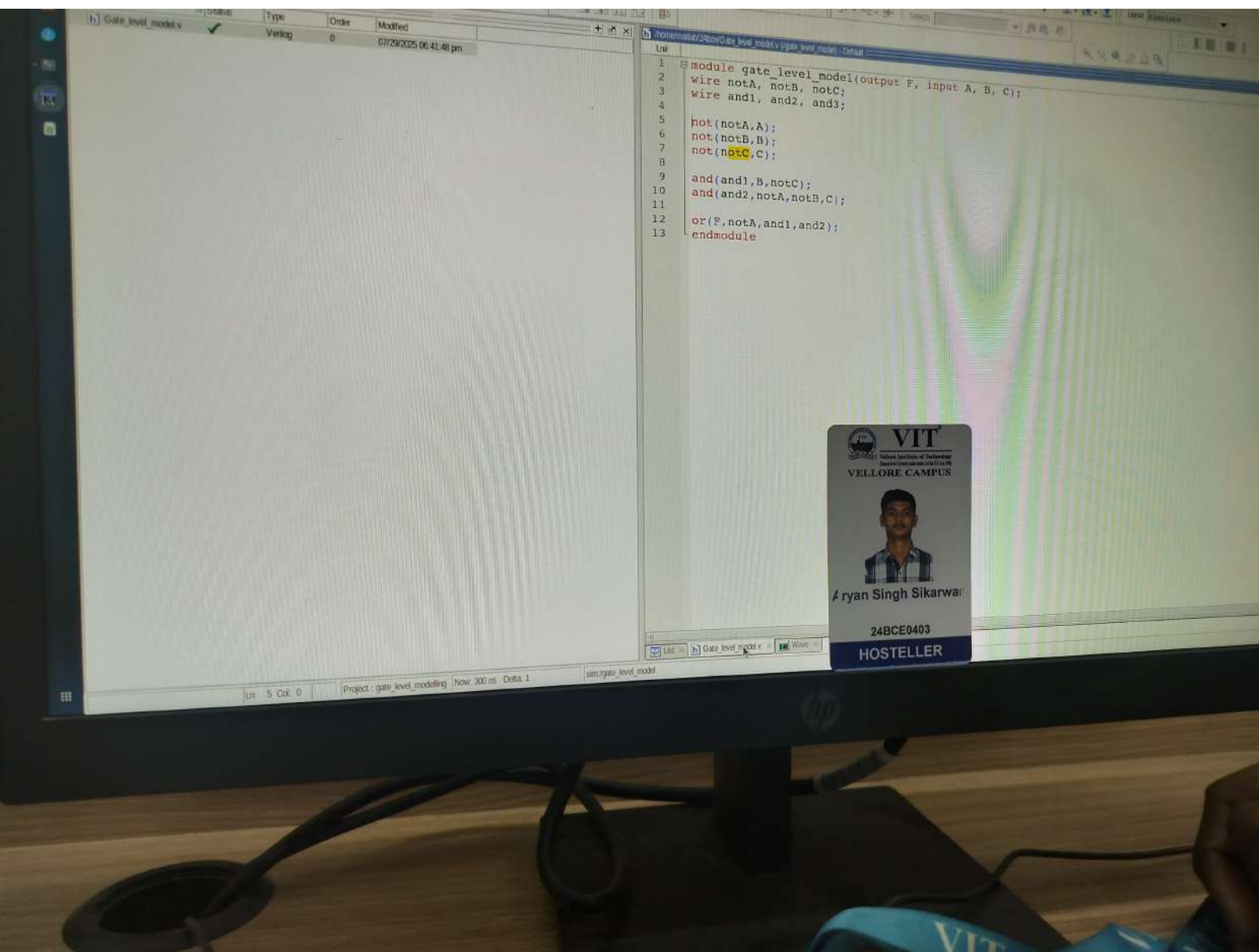
```
assign Y_not = !A;
```

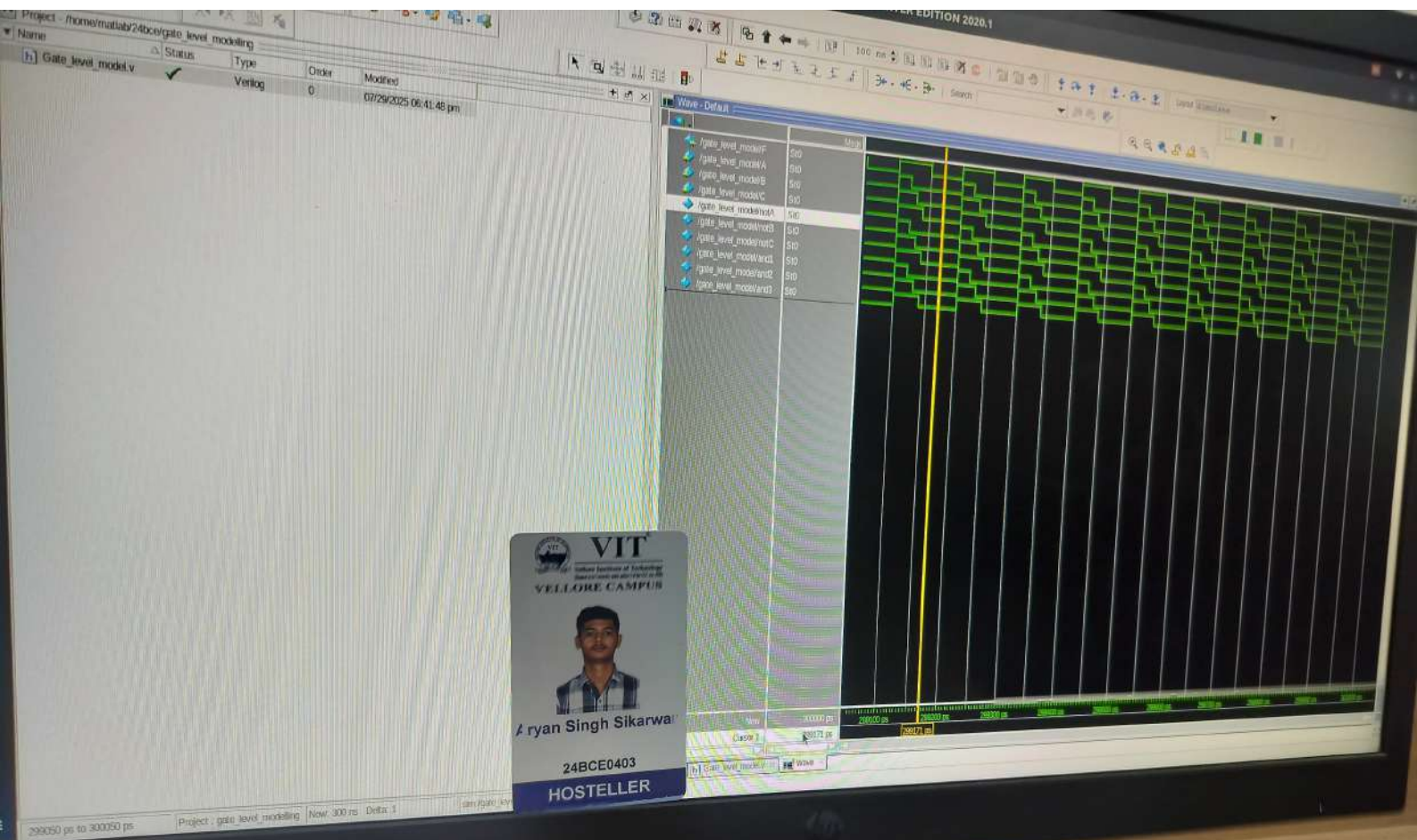
```
assign Y_nand = !(A & B);
```

```
assign Y_nor = !(A | B);
```

```
endmodule;
```







* problem 2 :-

=> write Verilog code in one file for all gates using Data flow modeling and then verify its operation. two level modeling.

```
code => module two-level (Y, A, B, C, D);  
    Input A, B, C, D;  
    Output Y;  
    time t1, t2;  
  
    and and-1 (t1, A, B);  
    and and-2 (t2, A, D);  
    or or-1 (t1, t2);  
end module
```

* Data flow modeling :- Q1

```
=> module AND-2-data-flow  
    (output Y, output A, B);  
    assign Y = A & B;  
end module
```

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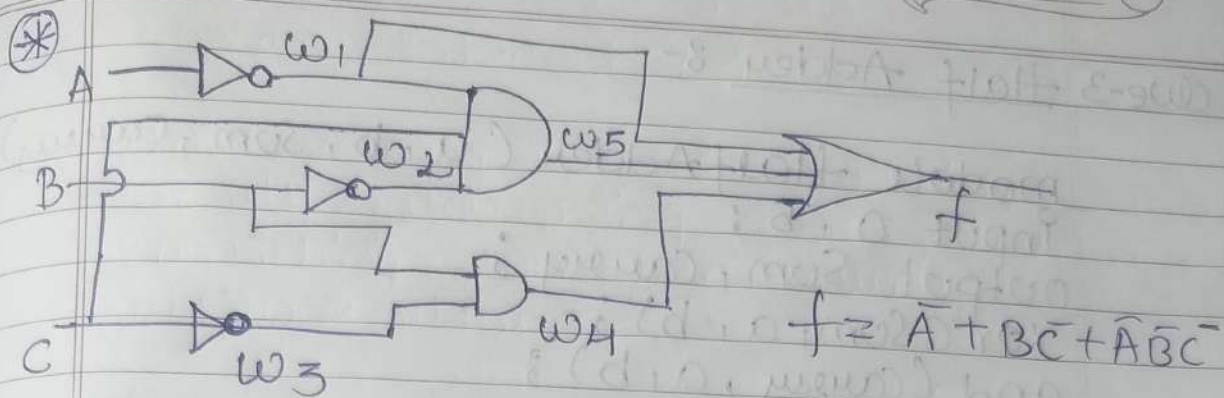
* Data flow modeling :- Q2

```
module data_flow_F (output F, input A, B, C);  
    assign F = !A | (B & !C) | (!A & B & C);  
end module
```

(Que-2)

classmate

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⇒ module ex (f, A, B, C);
Input A, B, C;
output f;

Wire w1, w2, w3, w4, w5;
not m0 (w1, A);
not m1 (w2, B);
and m2 (w3, C);

and m3 (w4, B, w1);
and m4 (w5, w1, C, w4);
or m5 (f, w5, w4, w4);
endmodule

A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$B\bar{C}$	$\bar{A}BC$	$f = A + B\bar{C} + \bar{A}BC$
0	0	0	1	1	1	0	0	1
0	0	1	1	1	0	0	1	1
0	1	0	1	0	1	1	0	1
0	1	1	1	0	0	0	0	1
1	0	0	0	1	1	0	0	0
1	0	1	0	1	0	0	0	0
1	1	0	0	0	1	1	0	1
1	1	1	0	0	0	0	0	0

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OIP verified

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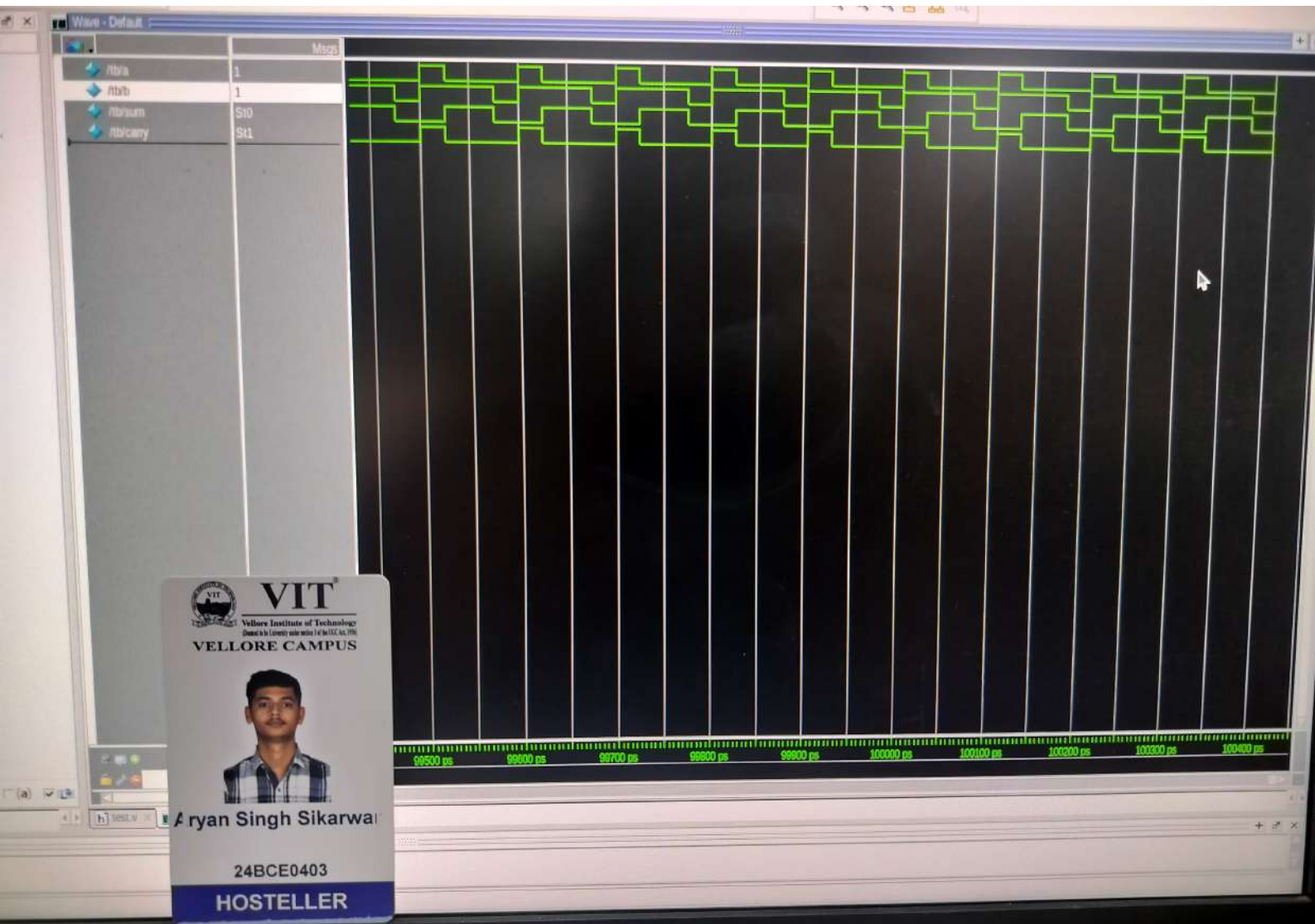
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```

1  module HalfAdder(a,b,sum,carry);
2      input a,b;
3      output sum,carry;
4      xor(sum,a,b);
5      and(carry,a,b);
6  endmodule
7  module tb();
8      reg a,b;
9      wire sum,carry;
10     HalfAdder mo(a,b,sum,carry);
11     initial
12     begin
13         a=1'b0;b=1'b0;
14         #100 a= 1'b0;b=1'b1;
15         #100 a = 1'b1;b=1'b0;
16         #100 a = 1'b1;b=1'b1;
17     end
18 endmodule
19

```





Ques-3 Half Adder :-

```

module HalfAdder (a, b, Sum, Carry);
    Input a, b;
    output Sum, Carry;
    xor (Sum, a, b);
    and (Carry, a, b);
endmodule

module tb();
    reg a, b;
    wire Sum, Carry;
    HalfAdder ma(a, b, Sum, Carry);
    initial
    begin
        a = 1'b0; b = 1'b0;
        #100 a = 1'b0; b = 1'b1;
        #100 a = 1'b1; b = 1'b0;
        #100 a = 1'b1; b = 1'b1;
    end
endmodule

```

2) Truth Table :-

A	B	Sum(s)	Carry (c)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

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 O/P Verified

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```

1 module Halfsubtractor(a,b,difference,borrow);
2   input a,b;
3   output difference,borrow;
4   xor(difference,a,b);
5   assign borrow=(-a&b);
6 endmodule
7 module tb_s();
8   reg a,b;
9   wire difference,borrow;
10  Halfsubtractor mo(a,b,difference,borrow);
11  initial
12  begin
13    a=1'b0;b=1'b0;
14    #100 a= 1'b0;b=1'b1;
15    #100 a = 1'b1;b=1'b0;
16    #100 a = 1'b1;b=1'b1;
17  end
18 endmodule
19
20

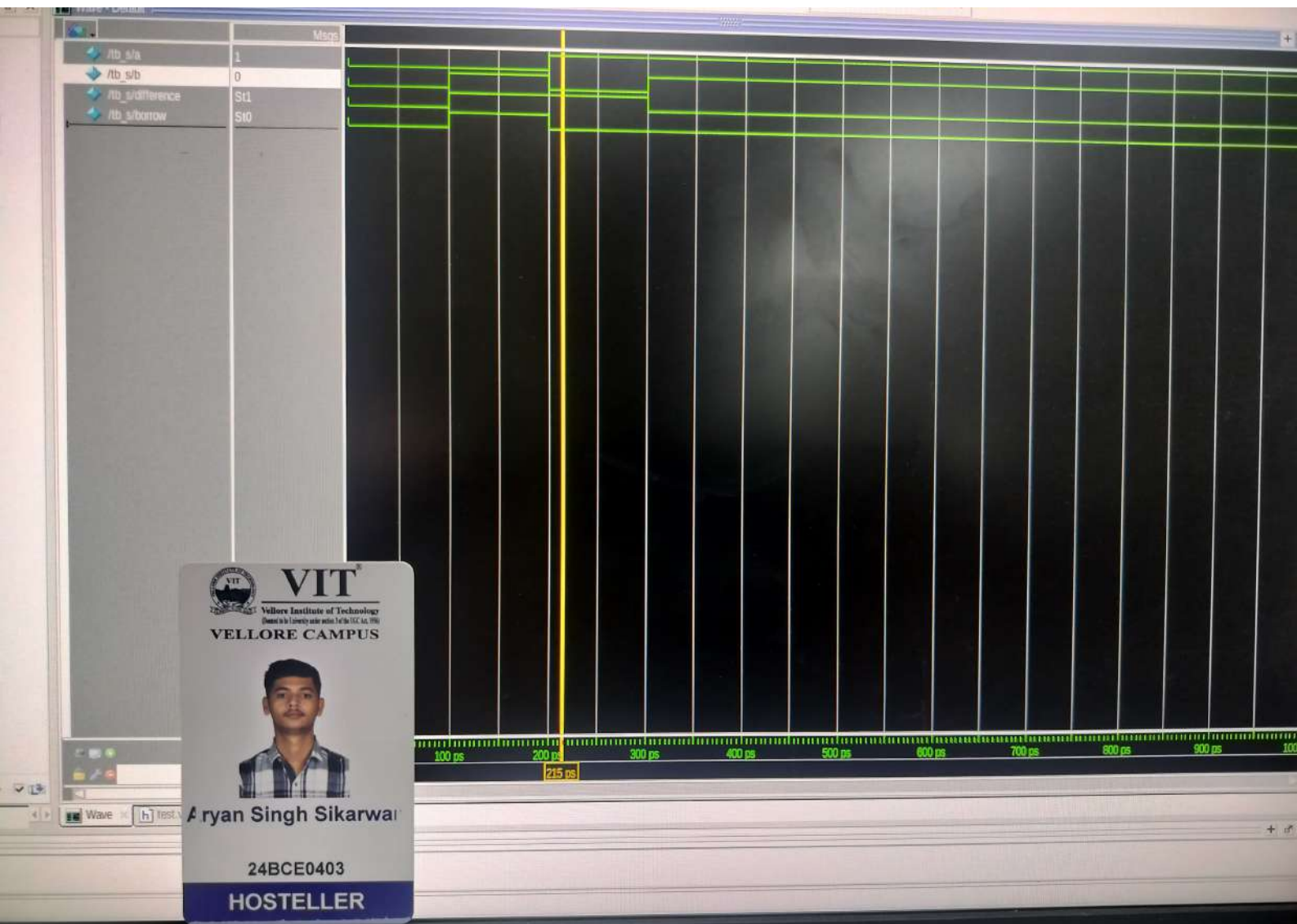
```


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Aryan Singh Sikarwal

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Que-2, part (B) :- Half Subtractor :-

1) Create Level :-

```
module half_subtractor (diff, borrow, A, B);  
    input A, B;  
    output diff, borrow;  
    wire not A;  
    xor (diff, A, B);  
    not (not A, A);  
    and (borrow, not A, B);  
endmodule
```

2) Data flow :-

```
module half_subtractor (output diff, borrow,  
    input A, B);  
    assign diff = A ^ B;  
    assign borrow = !A & B;  
endmodule
```

3) Truth table :-

A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0